Hedera Flow

The Decentralized Truth Layer for Utility Billing

Tokenizing Trust at the Grid Edge

Founder: NWAJARI EMMANUEL (De_real_iManuel), Nigeria | **Track:**

Technology Stack: Python, Google Vision AI OCR, Hedera HCS/HTS/EVM, AES-256 Encryption

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Electricity Billing Has a $96B Transparency Problem

Across emerging markets, electricity utilities lose over **$96 billion annually** due to billing errors, theft, and non-technical losses.

Millions of consumers receive **estimated electricity bills** instead of real meter readings.

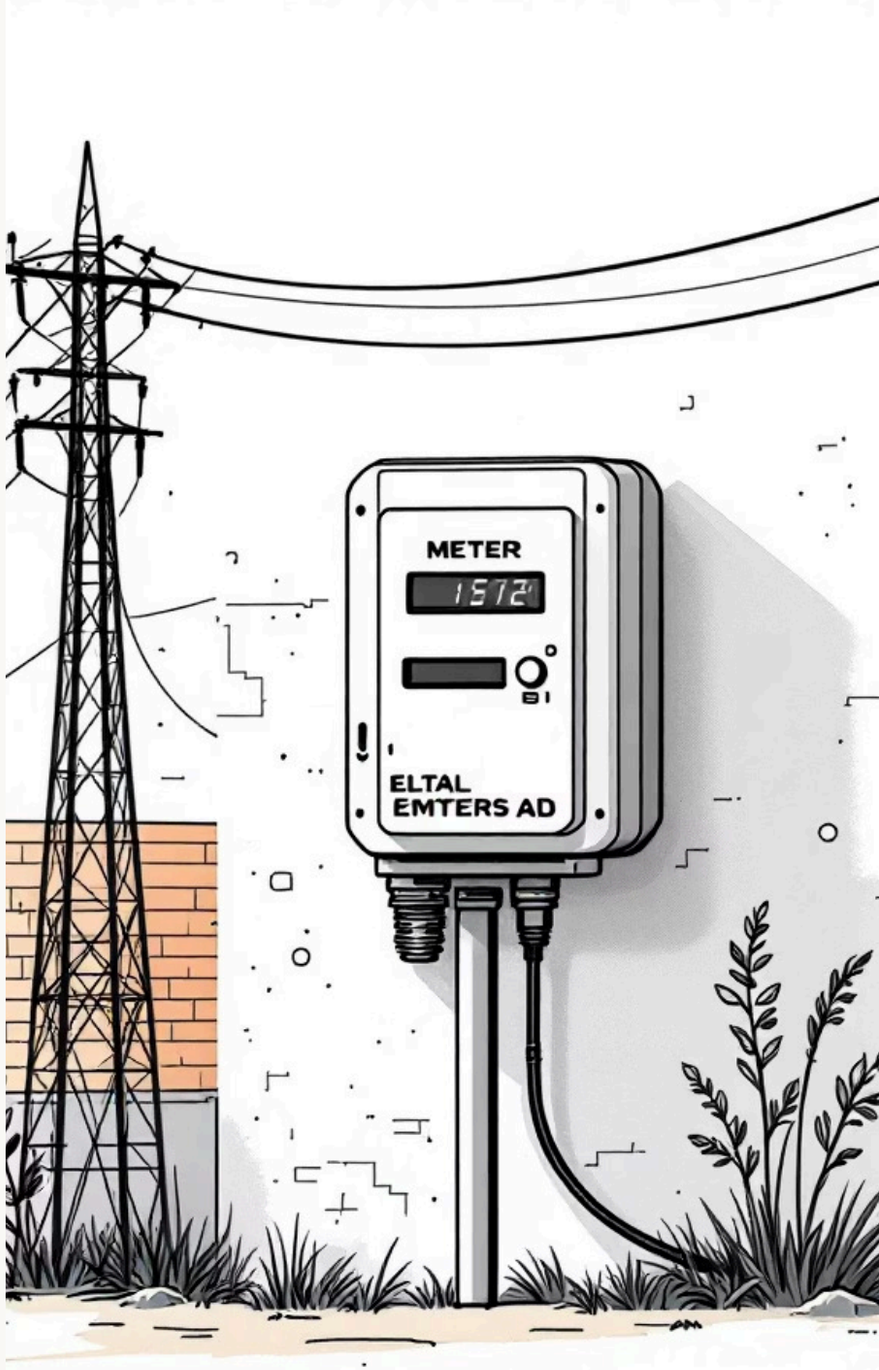
When Disputes Happen

- Consumers lack verifiable proof
- Utilities lack trusted data
- Billing disputes remain unresolved

Result

Trust between utilities and consumers breaks down.

Real User Story: Mrs. Adaeze, a teacher in Port Harcourt, saw her electricity bill suddenly triple despite no change in usage. She cannot prove the bill is wrong. The utility cannot verify the meter reading.



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Introducing Hedera Flow

A decentralized verification layer for electricity billing that creates a **cryptographically secure pipeline from physical electricity meters to an immutable blockchain ledger.**

1

Capture

AI-OCR extracts electricity meter readings using Google Vision and Tesseract

2

Secure

Consumer data protected with AES-256 encrypted local keys

3

Notarize

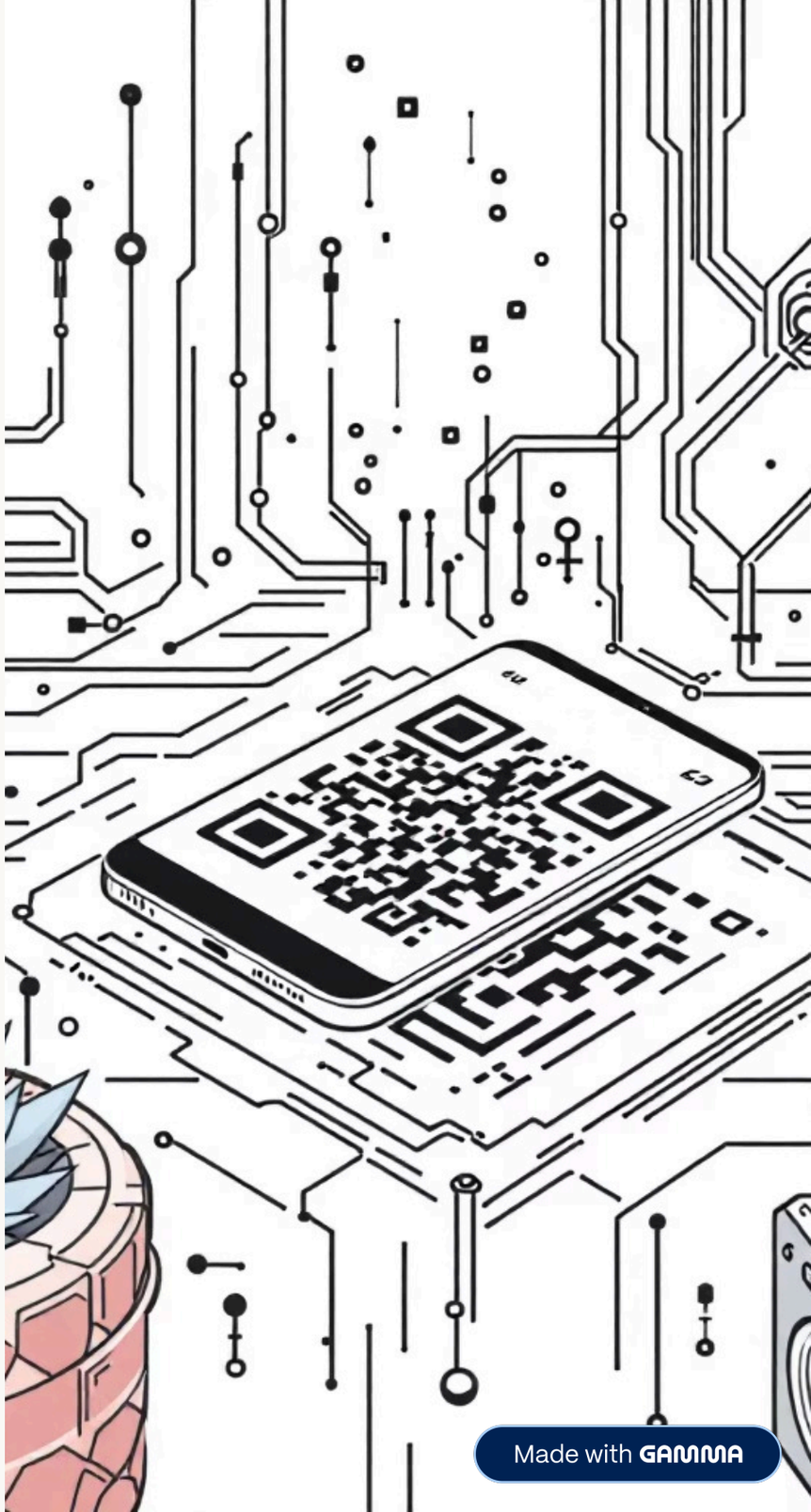
Meter readings hashed and submitted to the Hedera Consensus Service

4

Settle

Smart contracts automate billing through the Hedera EVM

This removes **human estimation and billing manipulation**.



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Built for Global Utility Scale

Instead of storing all readings in one stream, Hedera Flow uses **regional consensus topics** for scalability and compliance.

Lower Latency

Regional data processing

Regional Data Compliance

Local regulatory adherence

High Throughput Infrastructure

Massive scale support

Live Testnet Topics

Africa

0.0.8052391

Europe

0.0.8052384

United States

0.0.8052396

Asia

0.0.8052389

Each electricity reading becomes an **immutable timestamped message on the Hedera network.**

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From OCR to Device Identity

Hedera Flow is evolving from software-based verification to **device-level trust**, inspired by the machine identity model used by Sealcoin.

1	2
Current System • Hybrid OCR extracts electricity meter readings • Data is processed and anchored to Hedera	**Next Evolution** Smart meters sign readings directly using **device cryptographic keys**

1

Ed25519 Digital Signatures

2

Unique Cryptographic Identity Per Meter

> **Meter → Signed Reading → Hedera Network**
>
> No human intervention • No estimation • No manipulation

This creates a **trusted DePIN infrastructure for energy data.**

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Early Interest from Nigeria's Power Sector

Port Harcourt Electricity Distribution Plc

Feedback Received:

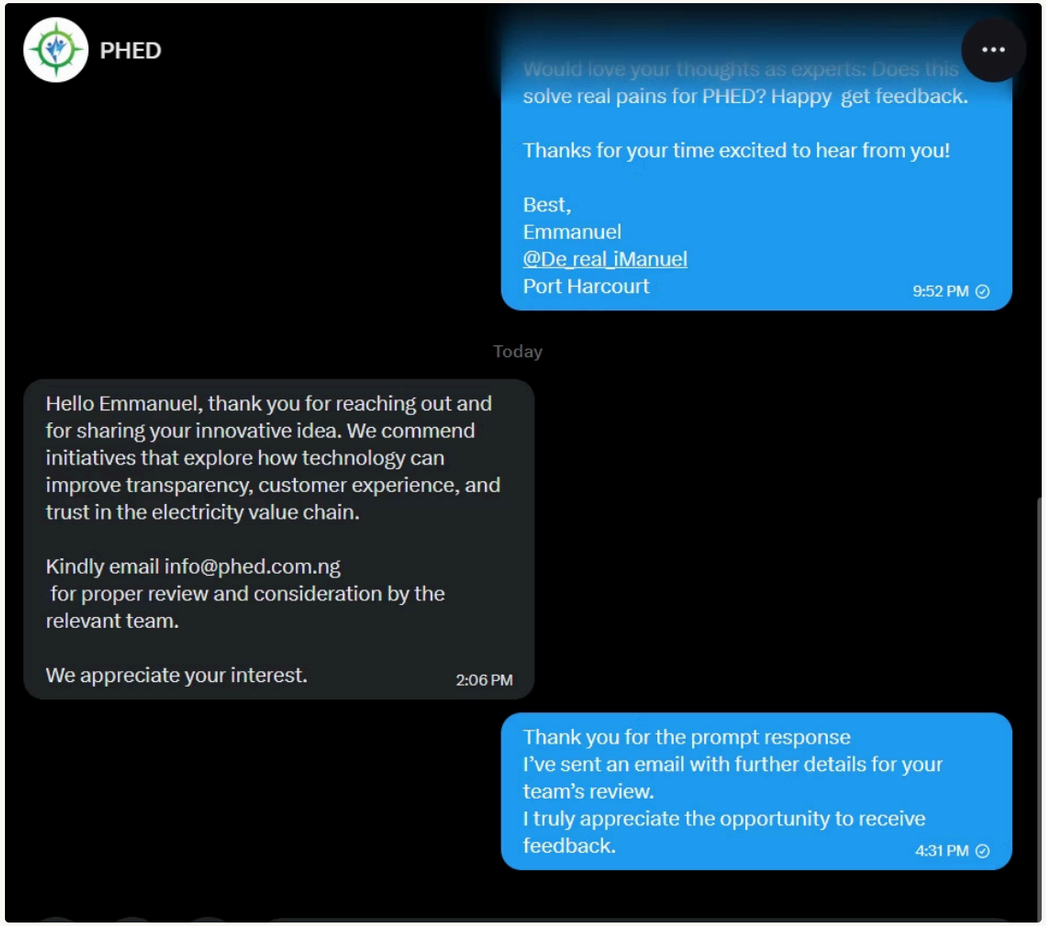
"We commend initiatives that explore how technology can improve transparency and trust in electricity billing."

Current Status

Internal technical review is ongoing

Goal

Launch pilot testing with real consumers in Port Harcourt



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Enterprise Infrastructure for Critical Utilities

Hedera provides the reliability required for national infrastructure systems.

Ultra-Low Fees

$0.0001 per consensus message

Security

Asynchronous Byzantine Fault Tolerance (ABFT)

Smart Contracts

Solidity compatibility through Hedera EVM

Tokenization

Energy credits issued via Hedera Token Service

Fast, scalable, and environmentally efficient.

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Monetizing Trust and Transparency

Hedera Flow generates revenue through multiple channels.

Utility SaaS Platform

Analytics dashboards that identify electricity theft and revenue leakage

Transaction Fees

Small convenience fees on verified electricity bill payments

Regulatory Data Services

Transparency reports for national regulators and grid audits

Unit Economics

- Software-first infrastructure
- No new hardware required for legacy meters

~98%

Estimated Gross Margin

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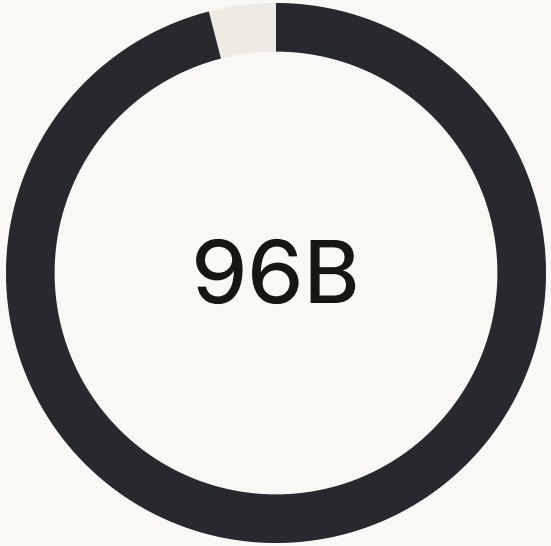
A Global Energy Infrastructure Opportunity

Electricity billing systems serve **billions of consumers globally.**

The smart meter and digital utility infrastructure market is rapidly expanding.

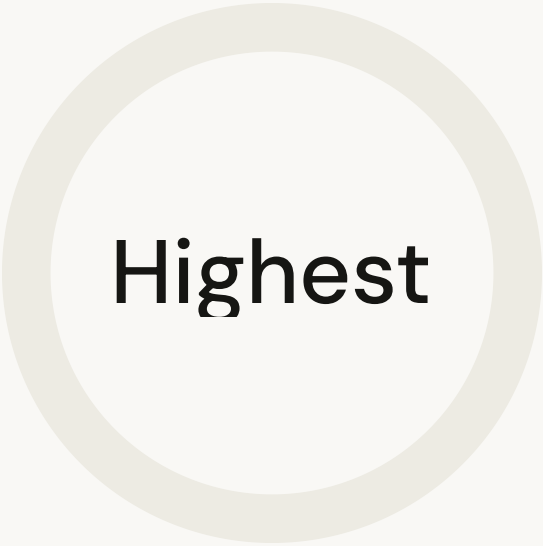


Annual Losses

Due to billing inefficiencies

Global Coverage

Billions of consumers served



Emerging Markets

Experience highest trust gaps

Hedera Flow can become the **verification layer for global electricity billing systems.**

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From Local Pilot to Global Infrastructure

1 — 2026

Launch **500-user pilot in Port Harcourt** with local utility collaboration

2 — Late 2026

Integrate **omnichain payment support** via ZetaChain. Users will be able to pay electricity bills using BTC, ETH, USDT. Payments settle instantly on Hedera

3 — 2027

Release **Flow-Stick hardware adapter** for industrial electricity meters

Building a Decentralized Energy Data Infrastructure Network

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